

Title: Developing a Linear Regression Model for Predicting Medical Aid

Charges

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1.Introduction

The increasing cost of healthcare has led to a greater need to accurately forecast charges in medical insurance. This has resulted in a greater need to utilize predictive analytics tools to enable organisations to measure financial risk and improve decision-making processes. One of the fundamental supervised learning techniques is Linear Regression, where a dependent variable is predicted in relation to one or more independent variables. This model is commonly applied to forecast a dependent variable based on one or more independent variables, making it a popular choice for predictive analytics.

The purpose of this research is to create a Linear Regression model to forecast charges in medical insurance. Through the patterns identified in the dataset, the research aims to determine some of the factors affecting charges in healthcare and the efficiency of a regression-based predictive model.

2.Dataset Overview and Suitability

The dataset applied in this research is the Medical Insurance Dataset, consisting of 1,338 data points and seven variables. The dataset includes information on age, sex, BMI, number of children, smoking, region, and charges.

The dependent variable, charges, is continuous, making it appropriate for a regression-based model.

The dataset is appropriate for a Linear Regression model for a number of reasons. Firstly, a regression model is appropriate when a continuous dependent variable is applied, as is the case in this research. Secondly, the dataset has multiple independent variables, making it appropriate for a regression model. Thirdly, the dataset is appropriate because it has no missing values, thus no preprocessing is necessary.

3. Planning Analysis

A structured plan was designed before commencing the project to ensure a systematic and efficient approach.

Exploratory Data Analysis (EDA)

The first step in this phase was to comprehend the data provided, which was done through summary statistics and data visualization techniques. EDA is a significant step in data analysis, as it helps identify patterns in the data, which is critical in decision-making (Tukey, 1977).

Feature Selection

The second phase, which is feature selection, was designed with both statistical and domain knowledge in mind. Features with strong relationships with the target variable, such as smoking status and BMI, were expected to play a critical role in determining the outcome. Feature selection techniques, such as forward and backward selection, were planned for this phase, although all features would be used in this project.

Model Training

The third phase, which is model training, was designed with a Linear Regression model in mind, which is efficient in its usage and ease of understanding. The dataset would be split into two sets: training and testing, which would help in determining how well the model would generalize with new data. One-Hot Encoding would be used on categorical features.

Model Evaluation

The fourth phase, which is model evaluation, would include using metrics such as Mean Absolute Error, Root Mean Squared Error, and R-squared to determine how well the model performed.

Reporting

The final phase, which is reporting, would include a structured report on the findings, which would include visualizations, interpretations, and recommendations.

4. Exploratory Data Analysis (EDA)

4.1 Dataset Structure and Summary Statistics

An initial inspection of the dataset indicated that no missing values were present, and all variables were correctly formatted.

The numerical variables, including age, BMI, and medical charges, exhibited varying ranges that reflected natural differences among respondents. These variables were ready for direct use in modeling without further cleaning.

The categorical variables, such as sex and smoking status, required encoding to convert them into a numerical format suitable for machine learning algorithms. One-Hot Encoding was later applied to these variables to ensure proper representation in the model.

Table 1: A summary of the key variables

Variable	Count	Mean	Std. Dev.	Min	25%	Median	75%	Max
Age	1338	39.21	14.05	18.00	27.00	39.00	51.00	64.00
BMI	1338	30.66	6.10	15.96	26.30	30.40	34.69	53.13
Children	1338	1.09	1.21	0.00	0.00	1.00	2.00	5.00
Charges	1338	13,270.42	12,110.01	1121.87	4740.29	9382.03	16,639.91	63,770.43

Age: The age of the respondents varied from 18 to 64 years, and the average age was around 39 years. This implies that the data points were largely distributed among adults in early to mid-adulthood.

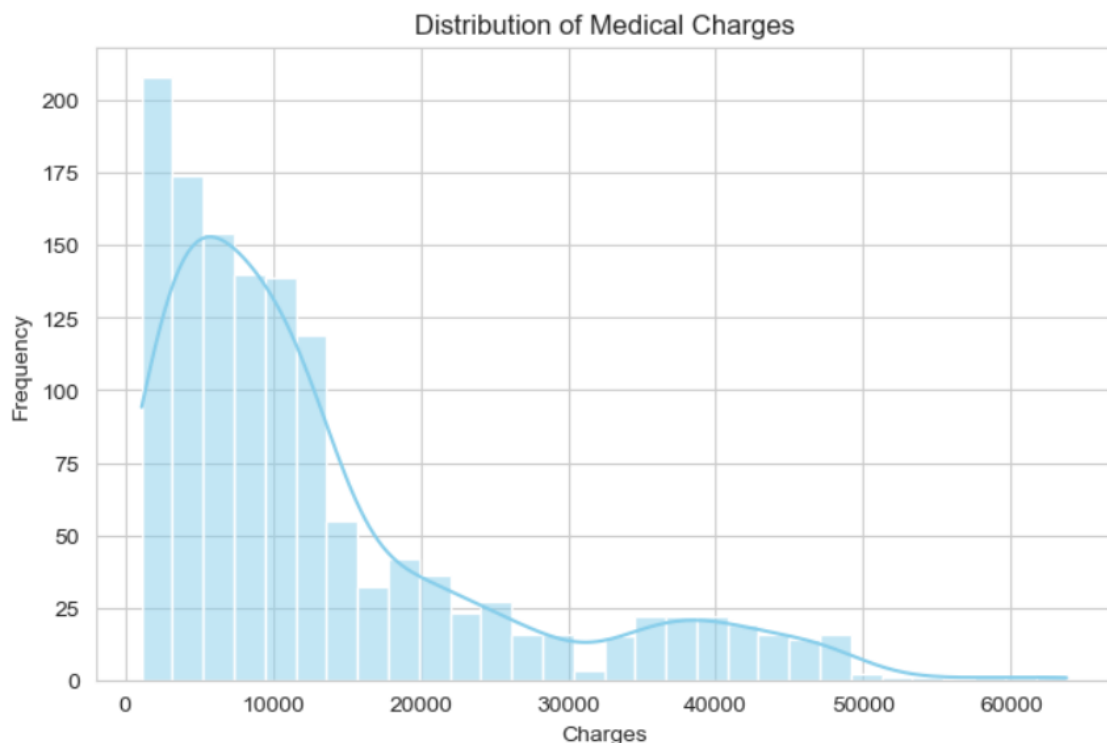
Body Mass Index (BMI): The average value of the body mass index for the population was around 30.7. This implies that a large proportion of the population were either overweight or obese.

Children: The majority of the population had between 0 and 2 children, and the average number of children per household was around 1.1. This implies that a large proportion of the population came from small to medium-sized families.

Medical Charges: The insurance charges for the population were heavily right-skewed, and the average value of these charges was around 13,270. The maximum value of these charges was even higher than 63,000. This implies that although a majority of the population incurred moderate charges, a small proportion of them incurred very high charges.

4.2 Distribution of Medical Charges

Figure 1: Distribution of Medical Charges heatmap



The histogram showed that most of the medical charges were within the lower range, i.e., between 0 and 10,000. The frequency decreased gradually with an increase in medical charges.

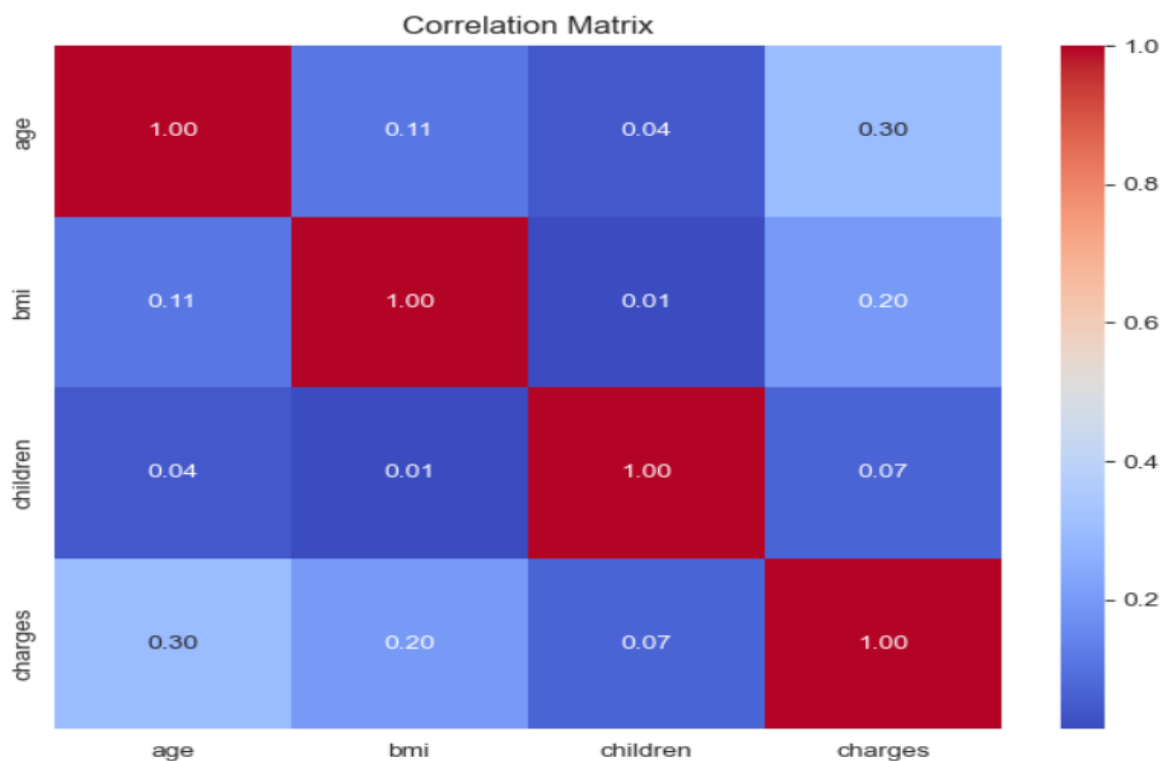
This showed a positively skewed distribution, with a small proportion of values showing higher medical charges, up to 60,000.

The results showed that healthcare expenditure was primarily driven by routine and low-cost cases, while a small proportion of higher cases were shown in the right tail. This was because these extreme values had a major impact on the distribution.

The median was considered to be a better average compared to the mean because the mean was thought to be affected by the presence of outliers, which did not accurately represent the average cost experienced by most patients.

4.3 Correlation Analysis

Figure 2: correlation Matrix



The correlation matrix revealed that the variable with the strongest positive relationship with medical charges was age, with a correlation coefficient of 0.30. This revealed that medical

charges tended to increase with age. BMI also had a positive relationship with medical charges, although the relationship was much weaker, with a correlation coefficient of 0.20.

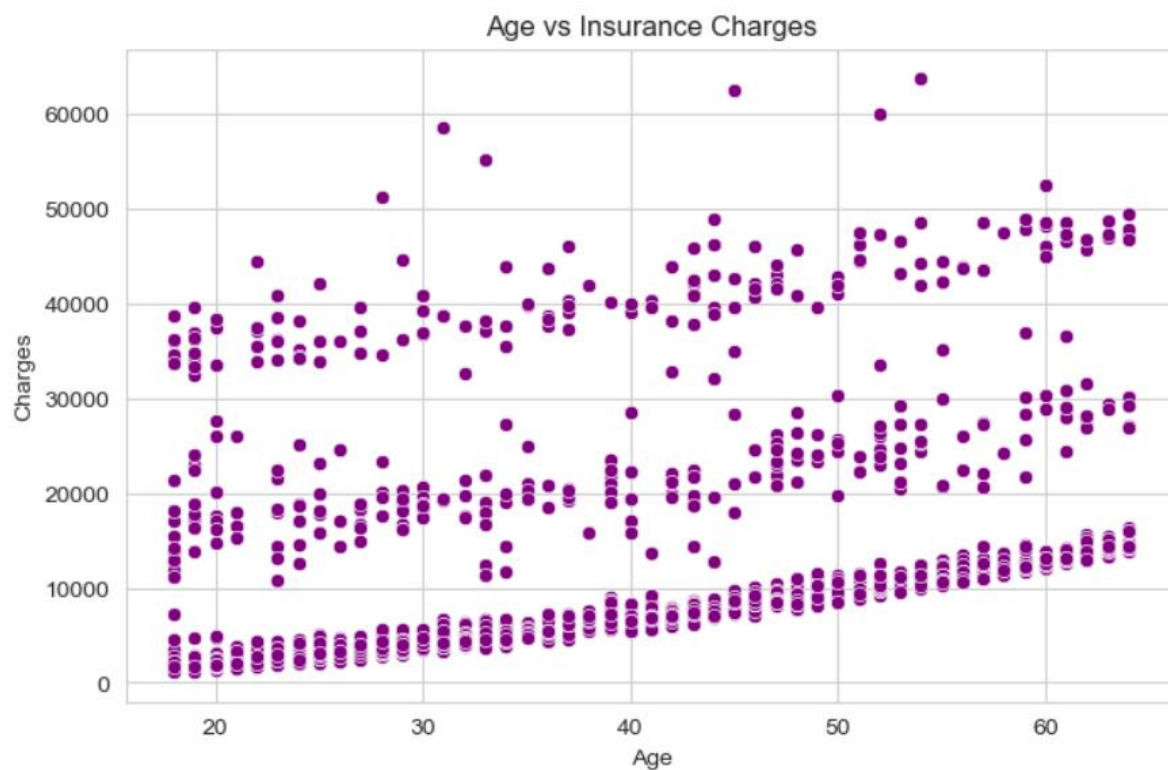
In contrast, the number of children had a very weak positive relationship with medical charges, with a correlation coefficient of 0.07. Moreover, the correlations between the independent variables were low, which meant there were no issues with multicollinearity.

Therefore, it can be concluded that the variable with the strongest impact on medical charges is age, followed by BMI, and then the number of children

This skewness in the data may result in a violation of the normality of residuals, which is a requirement in Linear Regression analysis (Osborne and Waters, 2002).

4.4 Relationship Between Variables

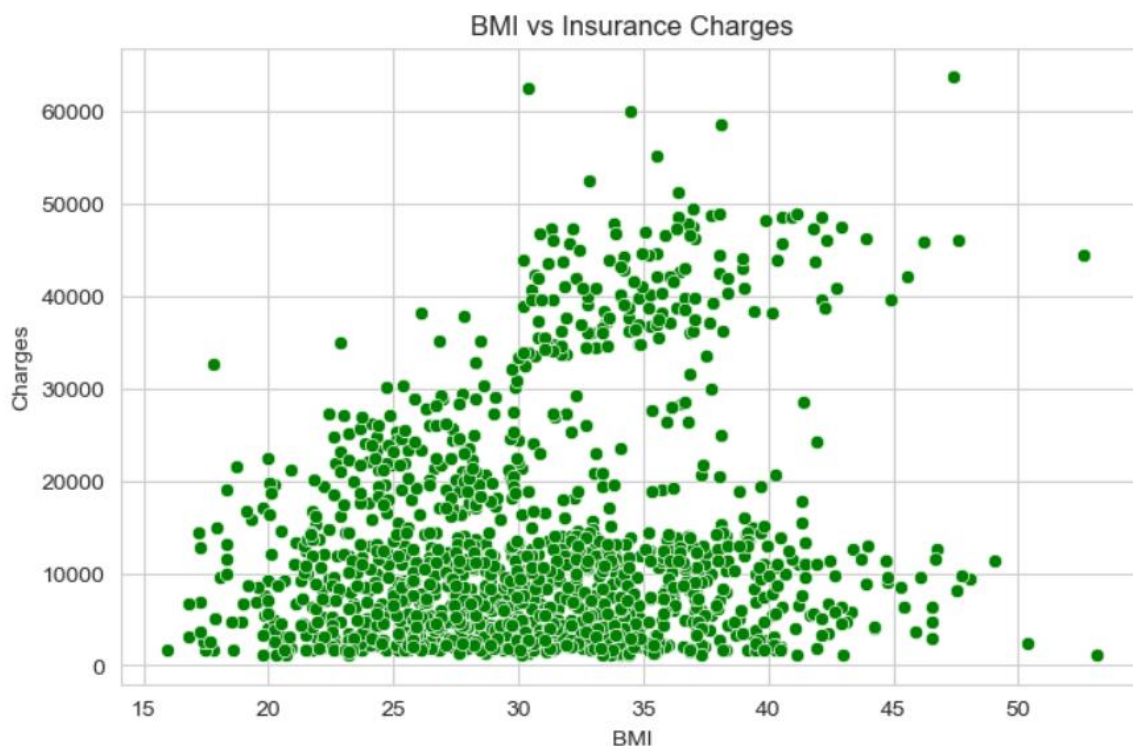
Figure 3: Age vs Insurance charges scatterplot



There is a positive relationship between age and medical charges, indicating a significant increase in healthcare costs with age.

This is in line with existing literature, which indicates that older people tend to seek more healthcare services.

Figure 4: BMI vs Insurance charges scatterplot

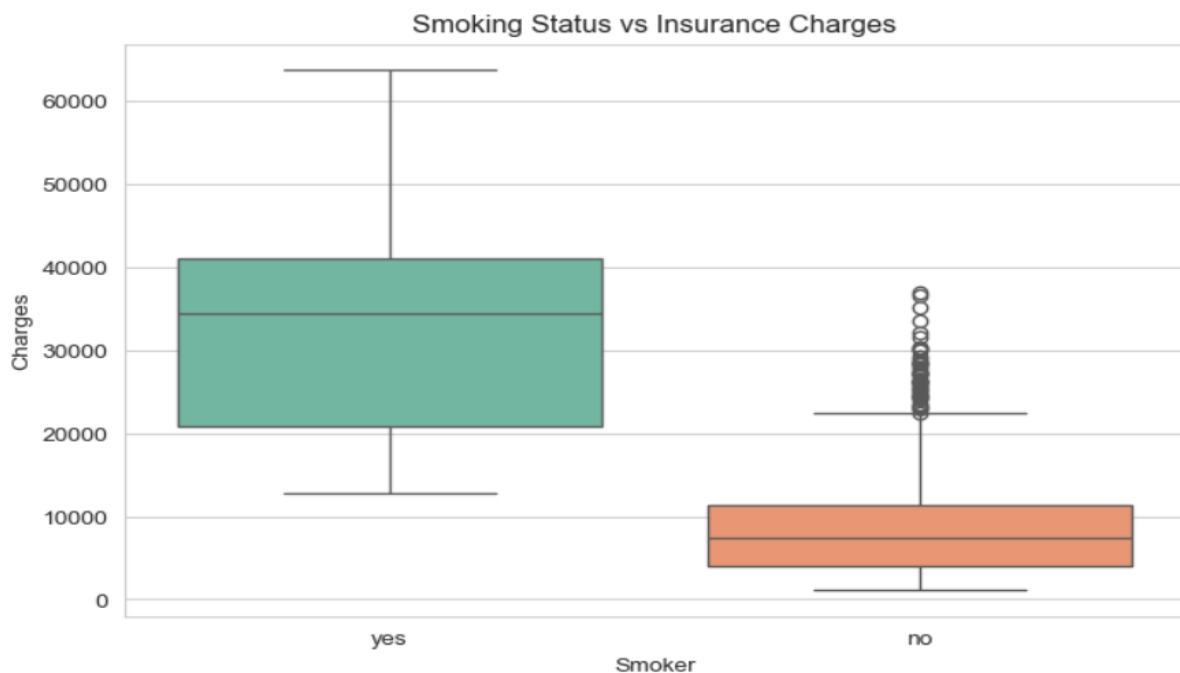


The scatter plot showed an overall positive relationship between Body Mass Index (BMI) and insurance charges. This means that as BMI increases, insurance charges also increase, but not in a perfectly linear fashion.

The points in the scatter plot were spread out, showing that there was more variability in insurance charges for higher values of BMI, indicating heteroscedasticity. While most individuals with BMIs between 25 and 35 had moderate insurance charges, below 20,000, there were some individuals with higher BMIs who had significantly higher insurance charges, creating noticeable upward outliers.

This showed that there was a positive relationship between higher BMIs and higher insurance charges, but it did not entirely explain insurance charges, which means that there might be other factors affecting insurance charges.

Figure 5: Smoking status vs insurance charges boxplot



The box plot helped to compare the charges for the smokers and non-smokers. The results showed that the smokers had a much higher median for the charges compared to the non-smokers. In addition, the range of charges for the smokers was much larger, with some charges going up to 60,000. On the other hand, the non-smokers had a much lower median for the charges. Moreover, the range of charges for the non-smokers was much smaller, although some of the charges were higher than 20,000. The results from the box plot showed a strong positive relationship between the charges for the smokers and non-smokers. All the data points for the smokers were much higher compared to those of the non-smokers. The implications of the results are that health charges for smokers are much higher, thus showing the importance of the factor in risk assessment.

4.5 Categorical Variable Analysis

The boxplots were used to analyze three different categorical variables: sex, region, and smoking.

Figure 6: Sex vs Charges boxplot

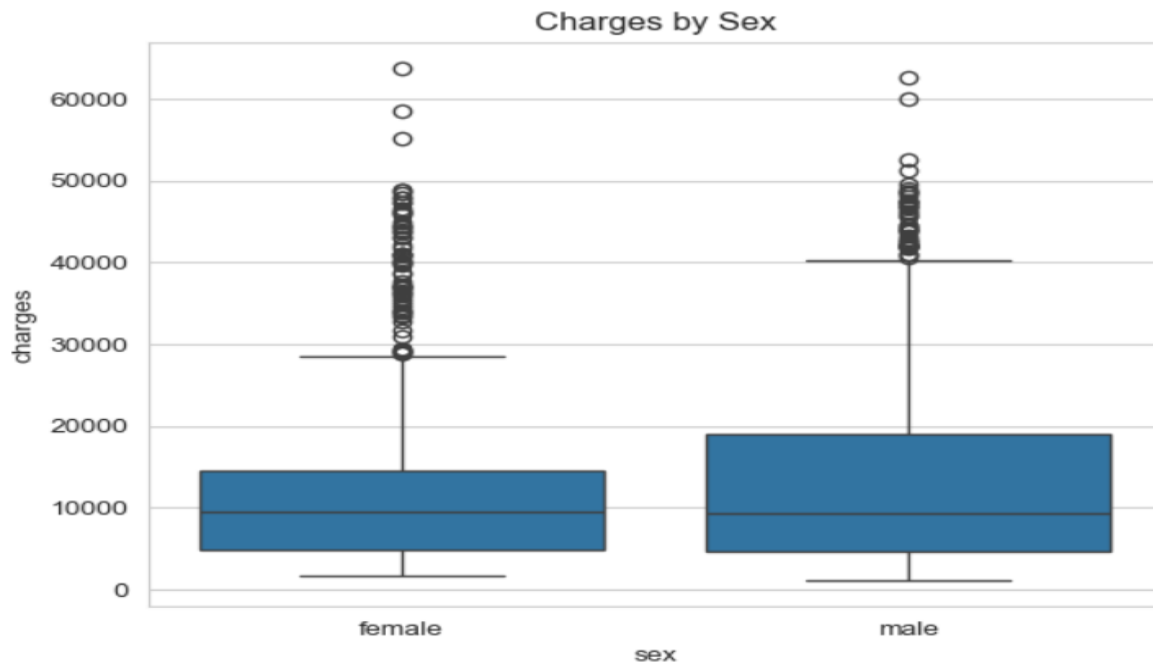


Figure 7: Region vs Charges

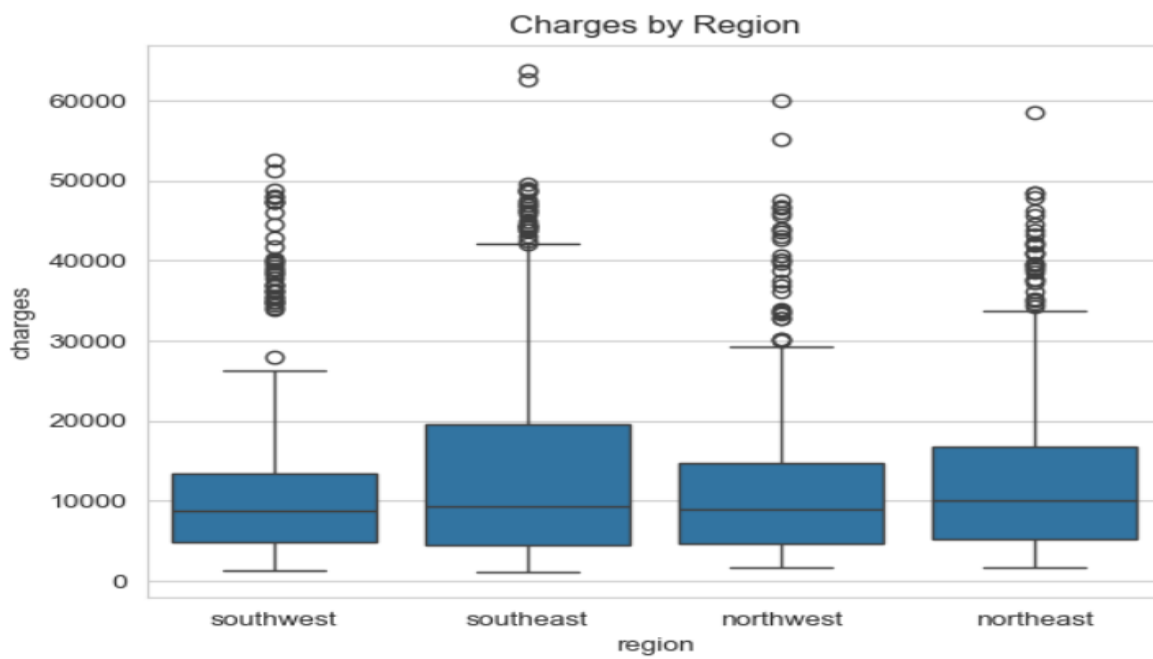
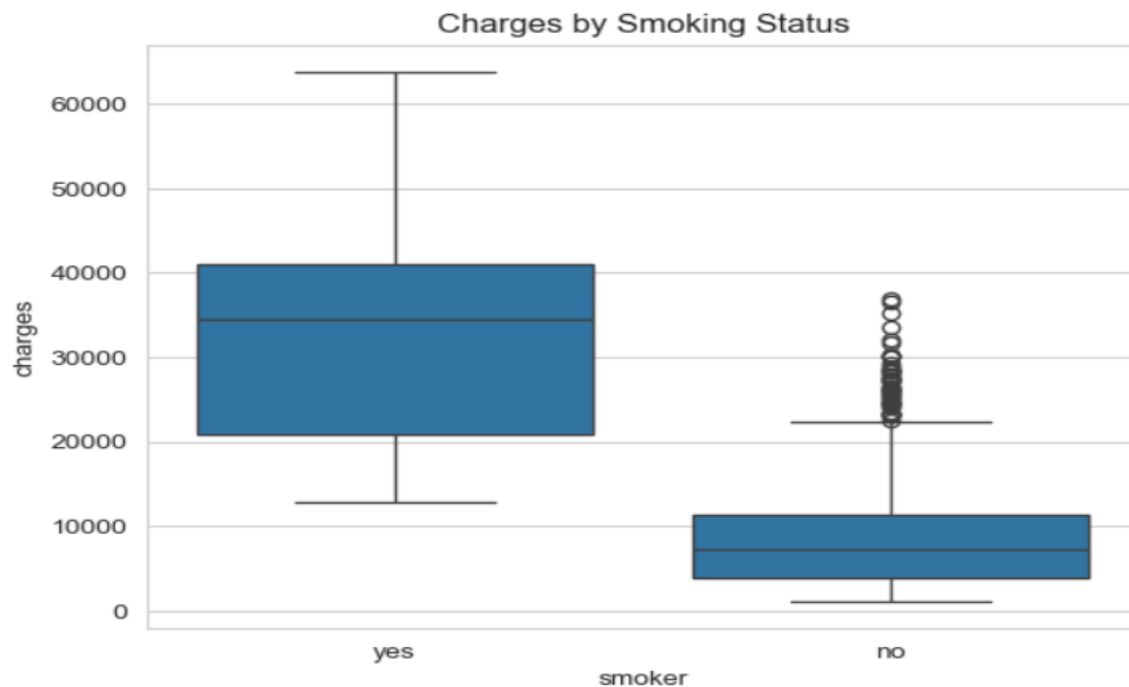


Figure 8: Smoking vs Charges boxplot



Interpretations of the above three figures:

Sex: The results showed a slight increase in median charge values for males compared to females.

Region: The southeast region showed a high median charge compared to other regions; however, there were some outliers in all regions.

Smoking: Smokers showed a high median charge of 30,000-35,000 compared to non-smokers' median charge of less than 10,000.

The results showed that the most significant factor affecting high medical costs is the smoker variable.

The results showed that insurance risk assessment should focus on the smoker variable as it always results in high charge values.

4.6 Key Insights from EDA

During the Exploratory Data Analysis (EDA), several key observations were made. The most significant factor affecting the level of charges incurred is the smoking status of a person. Smokers have significantly higher charges compared to those who do not smoke. The age and Body Mass Index (BMI) of a person also have a significant effect on the level of variability and charges incurred by a person. Finally, it is also observed that the charges are heavily skewed, and most of the patients have low charges incurred, while a small number of patients have very high charges incurred.

5. Data Processing

The categorical variables were one-hot encoded, which means that each category was converted to a binary variable. This method ensures that the Linear Regression model can process the categorical data in numerical form, without imposing an ordinal relationship that does not exist.

The numerical variables were used in their original form. This is because the Linear Regression model does not require any scaling of the features for prediction. The numerical values were used in their original form.

6. Feature Selection

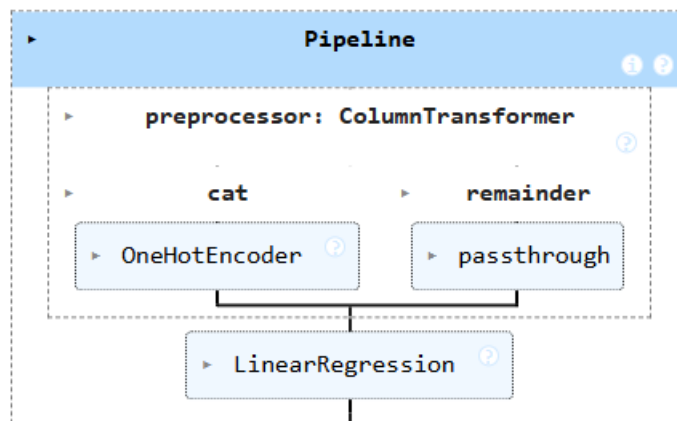
Feature selection was based on the results obtained in the exploratory data analysis (EDA) step. The results showed that the most significant features in determining insurance charges were smoking, BMI, and age. However, all features were included in the analysis to establish a comprehensive assessment of their potential contributions.

7. Model Training

The data set was split into a training set and a testing set, with 80% allocated as a training set and 20% as a testing set. This division enabled the model to learn from 80% of the data set and then make predictions on the 20%.

The Linear Regression model has been trained by utilizing a pipeline. The pipeline combines the data preprocessing stage and the model training stage. The use of a pipeline helped in achieving consistency in handling categorical variables, numerical variables, and the model training. By utilizing a pipeline, the reproducibility of the code, consistency in the data flow, and minimization of errors have been achieved. Moreover, the use of a pipeline has allowed for flexibility in future changes.

Figure 9: model pipeline



8. Model Predictions

Predictions were made on the test dataset using the trained model. Based on input features, these forecasts showed projected medical costs.

9. Model Evaluation

Table 2: Model Performance Metrics

Metric	Value
MAE	5796.28
RSME	4181.19
R^2	0.78

The performance of the Linear Regression model has been measured by Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and R-squared (R^2) scores.

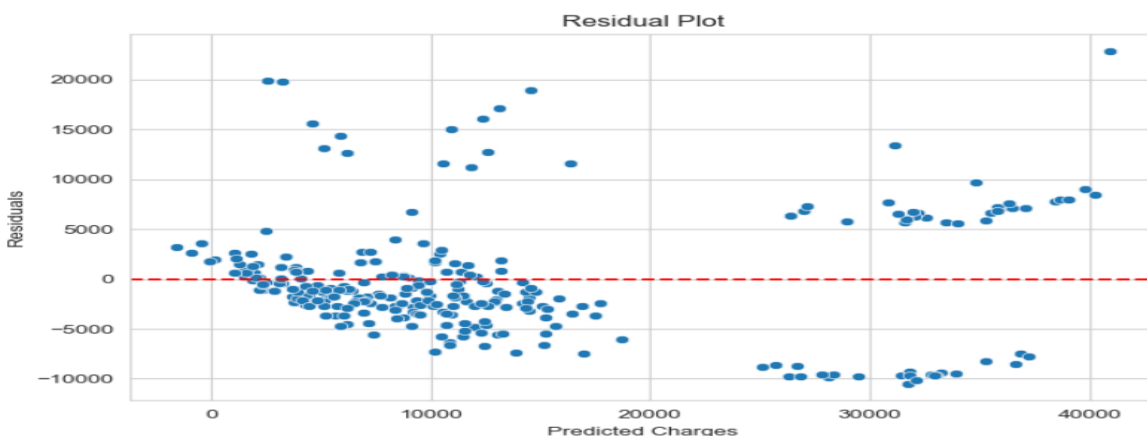
The RMSE of 5,796.28 indicated the square root of the average of the squared differences between the predicted values and the actual values. The high value of RMSE indicated that some of the predictions made by the model were far off from the actual values, considering the high cost of some of the patients.

The MAE of 4,181.19 indicated that on average, the predicted values made by the model were off by 4,181 units from the actual insurance charges. Thus, there is a need for improvement in the accuracy of the prediction made by the model.

The R^2 score of 0.78 indicated that the model has been able to explain 78% of the total variability in the medical insurance charges. Thus, the performance of the model has been satisfactory in terms of prediction accuracy, as most of the variables have been captured by the model. However, there is a need for improvement in the accuracy of the prediction made by the model because 22% of the variables have not been explained by the model. Thus, there are some other factors that influence the cost of the medical insurance charges, which have not been captured by the model.

9.1 Residual Analysis

Figure 10: Residual Analysis scatterplot



The residual plot is used to demonstrate the discrepancies between the actual values and the predicted values generated by the Linear Regression model for the medical charges. The residuals were plotted against the predicted values, with the red dashed line indicating perfect predictions.

Key Observations:

Centered Residual Plot: The majority of the residuals were close to the zero line, indicating that the predictions made by the model were accurate.

Spread of Residuals: The residuals were dispersed as the predicted charges increased.

Heteroscedasticity: The dispersed residuals at higher predicted charges indicate potential heteroscedasticity, implying that the prediction errors were not uniform for all medical charges.

No Strong Nonlinear Pattern: The residuals did not demonstrate a strong curve, indicating the appropriateness of the linear model. However, the high values were difficult to predict.

Interpretation:

The performance of the model is reasonable, as the residuals are generally balanced on either side of the zero line. However, the accuracy of the predictions is reduced at high medical charges, possibly due to the high skewness in the data. The Linear Regression model performed well in modeling the relationships between the variables; however, other approaches, such as tree-based methods, might be more effective in dealing with the high variability observed in the data.

10. Discussion

The results showed that lifestyle and health-related factors had a major impact on medical insurance costs. Smoking had the biggest effect, followed by age and BMI. These results were in line with previous studies on factors that influence healthcare costs.

The skewed distribution of charges and possible non-linear relationships limited the effectiveness of the Linear Regression model, despite its good performance.

11. Model improvement

A number of enhancements were noted:

- Using a log transformation to lessen skewness
- Using regularization methods like Lasso or Ridge
- Investigating non-linear models like Random Forest Adding more variables

12. Conclusion

A Linear Regression model to forecast medical insurance costs was effectively developed by the study. Age, BMI, and smoking were found to be important cost determinants.

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